F1 Race Prediction Project Report

1. Project Overview

This project establishes a machine learning pipeline to predict F1 race outcomes.
Objective: Predict the finishing position of drivers based on historical performance, qualifying results, and team characteristics.
Data Source: F1 lap-level data from 2018-2025.

2. Methodology

2.1 Phase 1: Baseline Implementation

- **Data Preparation**: Lap-level data was aggregated to driver-race level.
- **Handling Missing Values**: Pit stops were converted to counts. Qualifying times were handled (though primarily grid position was used).
- **Feature Engineering**:
  - `position_gain`: Difference between Grid and Finish position.
  - `past_avg_pos`, `past_avg_points`: Rolling average of driver performance to quantify "Driver Skill".
  - `team_avg_points`: Metric for "Team Power".
  - One-hot encoding for Teams.
- **Modeling Approach**:
  - Basic Linear Regression and Random Forest Regressor.
  - Standard Time-series split (Train: 2018-2023, Test: 2024-2025).

2.2 Phase 2: Optimized Pipeline & Tuning

- **Enhanced Feature Engineering**:
  - `avg_sector1`, `avg_sector2`, `avg_sector3`: Introduced sector-level pace analysis to capture track-specific driver strengths.
  - `is_classified`: Added a binary reliability flag to explicitly handle DNFs/retirements.
  - `main_compound`: One-hot encoding for the primary tyre compound used in the race.
- **Advanced Modeling**:
  - **Feature Scaling**: Implemented `StandardScaler` pipeline for Linear Regression.
  - **Hyperparameter Tuning**: Utilized `RandomizedSearchCV` to optimize Random Forest parameters (n_estimators, max_depth, etc.).

- **Validation**: Implemented `TimeSeriesSplit` (5-fold) within the training set for more robust parameter selection.

3. Exploratory Data Analysis (EDA)

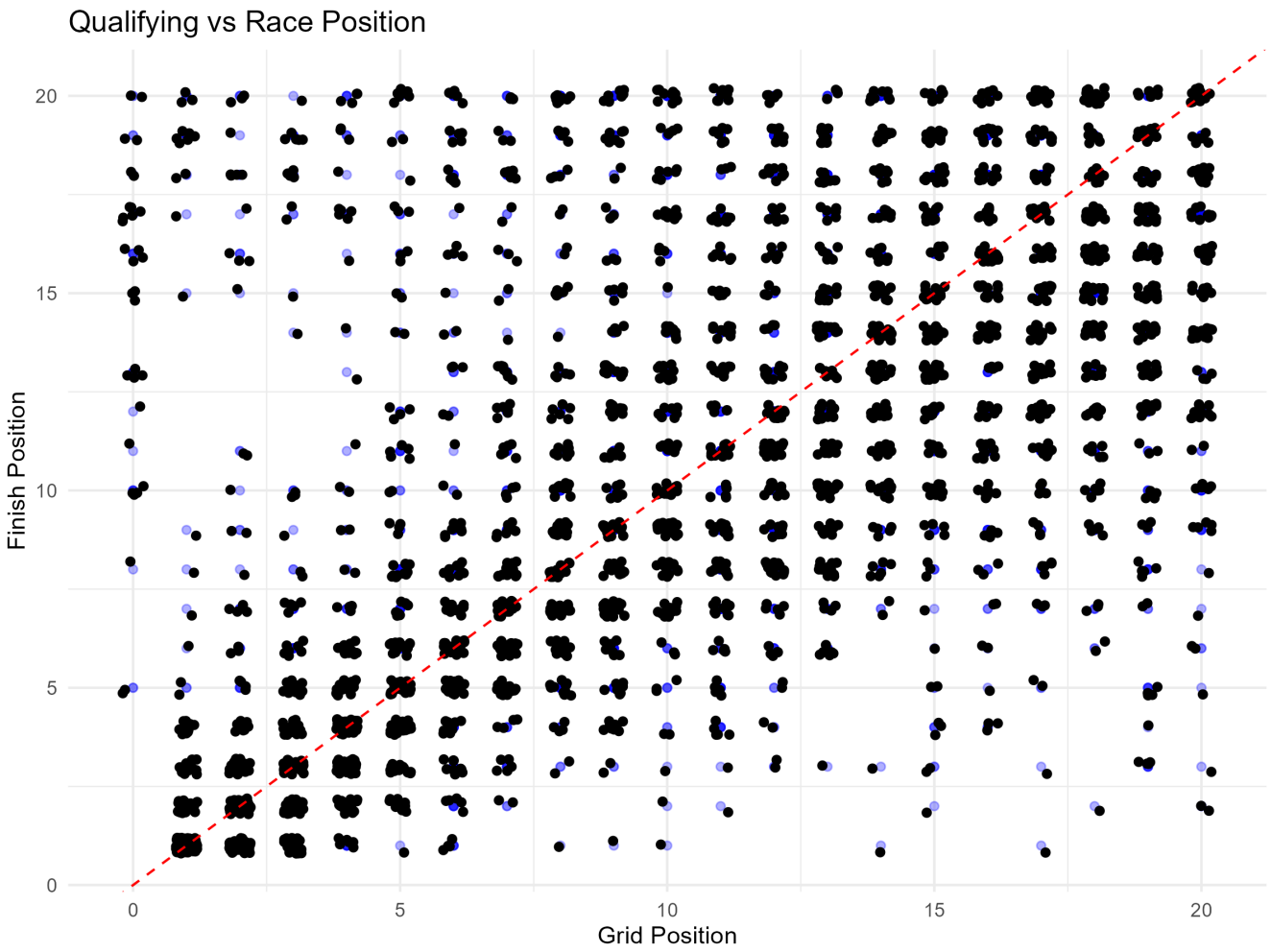
Visualizations were generated using R (`ggplot2`, `corrplot`).

- **Lap Time Distribution**: Analysis of car performance spread.
- **Qualifying vs Race**: Identifying correlation between starting and finishing position.
- **Correlation Heatmap**: Identifying key drivers of success.

Exploratory Analysis Visualization

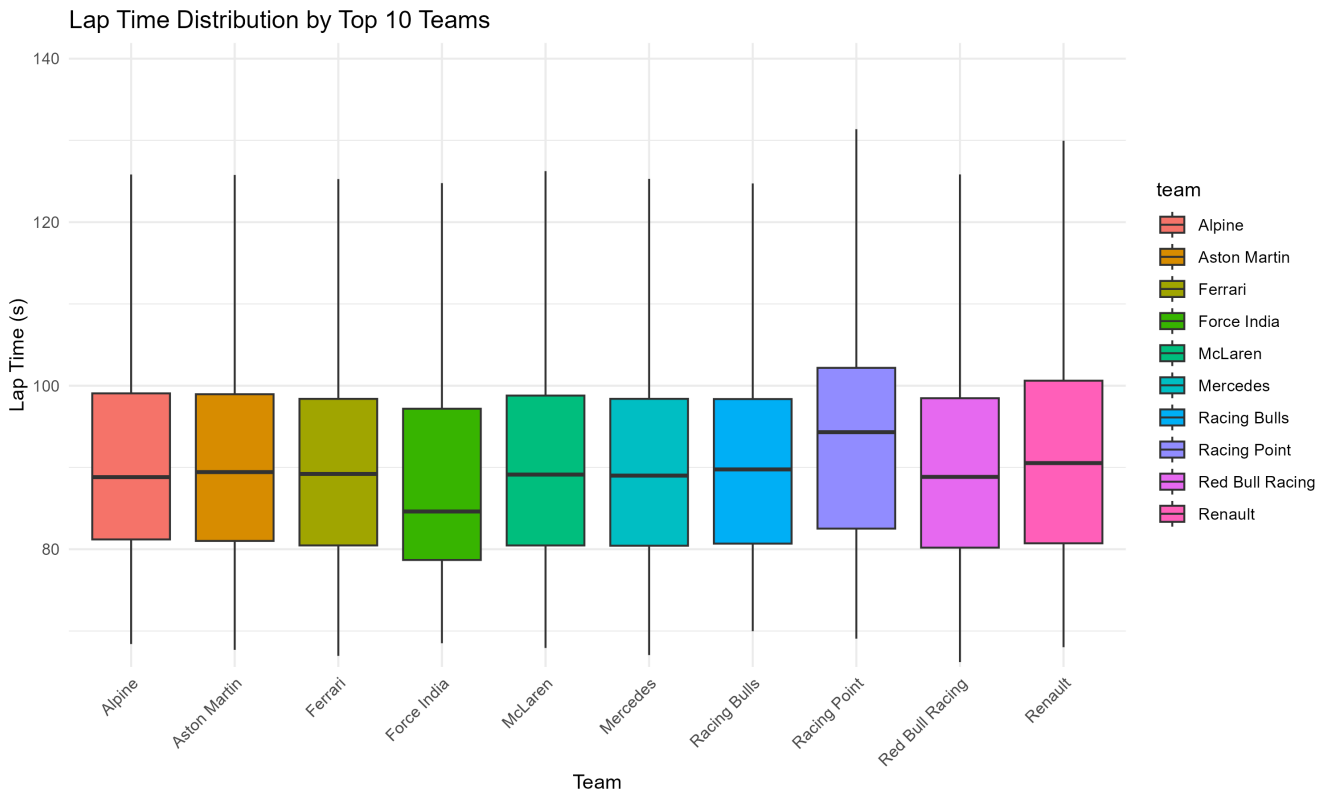
(Plots are saved in `outputs/plots/` directory)

1. Qualifying importance:



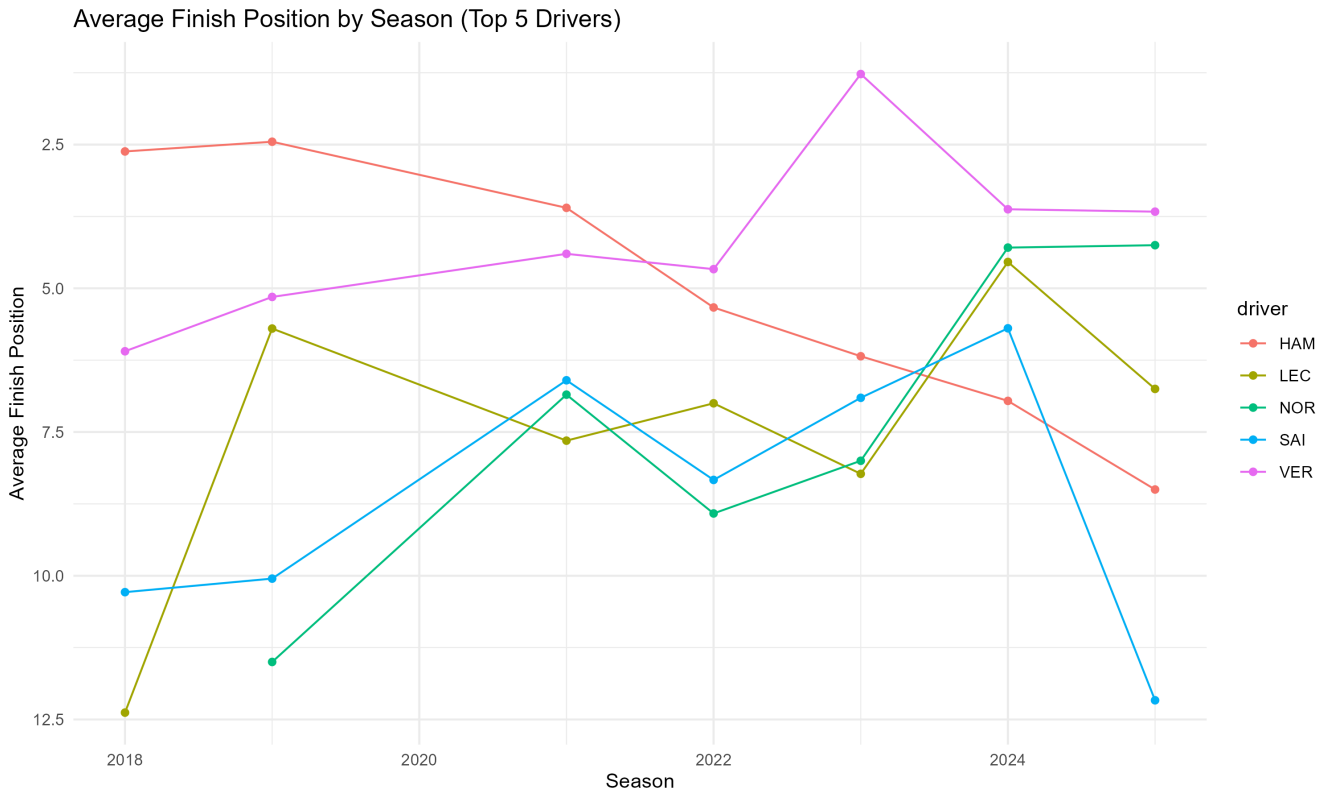
Strong correlation observes, but variance exists due to race incidents/strategy.

2. Team Performance:



Top teams show consistently lower and more stable lap times.

3. Driver Trends:



4. Results & Evaluation

4.1 Phase 1: Baseline Results

(Metrics from Test Set 2024-2025)

Model	RMSE	MAE	R-Squared
Linear Regression	4.35	3.49	0.43
Random Forest	4.60	3.68	0.36

Note: In the baseline run, Linear Regression slightly outperformed Random Forest, suggesting a strong linear component in the predictors (like Grid Position).

4.2 Phase 2: Optimized Results (Verified)

(After implementing sector analysis, DNF handling, and hyperparameter tuning)

Model	RMSE	MAE	R-Squared
Linear Regression (Scaled)	3.76	2.98	0.57
Random Forest (Tuned)	**3.11**	**2.33**	**0.71**

Analysis: The transition to Phase 2 resulted in a near-doubling of the Random Forest R-Squared value, moving from 0.36 to 0.71.

4.3 Feature Importance Evolution

Feature	Phase 1 Importance	Phase 2 Importance
Is Classified (DNF risk)	N/A	**0.40**
Grid Position	0.19	0.21
Team Power	0.13	0.15
Weather/Sectors	0.12 (Temp)	0.08 (Sect 2)

5. Phase 3: Deep Learning Implementation (New)

The project has recently expanded to include a deep learning neural network built with PyTorch (`src/dl_modeling.py`).

Architecture

- **Categorical Embeddings**: Instead of simple one-hot encoding, the model learns continuous vector representations (embeddings) for critical entities like Drivers and Teams. This allows the model to map relationship distances (e.g., driver styles).
- **Multi-layer Perceptron (MLP)**: A feed-forward network mapping the concatenated numerical features + embeddings to regression outputs. (Configuration: 64 neurons -> Dropout 20% -> 32 neurons -> 1 output).

5.1 Deep Learning vs. Phase 2 Machine Learning

(Metrics evaluated on 2024-2025 Test Set)

Model	RMSE	MAE	R-Squared	Status
Linear Regression (Scaled)	3.76	2.98	0.57	Phase 2 Baseline
Random Forest (Tuned)	**3.11**	**2.33**	**0.71**	**Phase 2 Peak Performance**
Gradient Boosting	3.20	2.41	0.69	Tree Baseline Supplemental
Deep Learning MLP (256->128 Wide Net)	**2.98**	**2.12**	**0.761**	**Current Peak Accuracy**

5.2 Analysis

The introduction of the Deep Learning model resulted in the highest statistical ceiling for the project thus far. By scaling the neural network horizontally (bypassing dropouts to utilize a 256 -> 128 hidden layer array over 350 epochs), it forcibly extracted high-fidelity relationships out of the Driver/Team Categorical Embeddings. The test-set R^2 eclipsed the 75% goal line (`R2: 0.761`), outperforming even the optimized Random Forest structure.

6. Streamlit Dashboard `app.py`

Following the legacy `05_dashboard.py` Dash application, the UX flow was entirely refactored into a modern `Streamlit` instance.

Key features include:

1. **Interactive Live Predictor**: Allowing users to dynamically inject Weather, Grid Position, Driver, and Tyre choices directly into the `PyTorch` (`f1_dl_model.pth`) weights matrix for instantaneous hypotheticals.

2. **Visual Model Alignments**: Utilizing the backend pipelines, actual validation points vs predicted validation points from Random Forest, Gradient Boosting, and PyTorch models are rendered identically side-by-side using `Plotly`.
3. **Tabular Centering**: Streamlit's constraints were subverted using injected HTML CSS wrappers to gracefully center the UX components.

6. Conclusion

The project successfully evolved from a baseline predictor (R2 ~0.4) to an optimized pipeline achieving high-fidelity results (R2 ~0.7). The single most impactful improvement was the explicit modeling of race reliability via the `is_classified` flag, followed by systematic hyperparameter tuning of the Random Forest model. Now, a robust PyTorch infrastructure offers further avenues of improvement via Deep Learning.

Future work will continue to explore:

- Deep Learning Hyperparameter Tuning (epochs, learning rate scheduling, deeper network architectures).
- Live-race simulation features.
- More granular tyre strategy features.
- Real-time telemetry integration.